

Indivisible Goods Recap

Definition 1. We have a set $N = [n]$ of players and a set $\mathcal{M}$ of m goods. Each player i has a nonnegative, additive valuation $v_i(S) = \sum_{g \in S} v_i(g)$. An allocation $A = (A_1, \dots, A_n)$ is a partition of $\mathcal{M}$ into disjoint bundles, one per player. Empty bundles are allowed; all goods are allocated.

For unnormalized values, proportionality then asks for $v_i(A_i) \geq v_i(\mathcal{M})/n$. Envy-freeness cannot always be achieved, so we have relaxed notions.

Definition 2. For an allocation $A = (A_1, \dots, A_n)$:

1. The allocation is *envy-free up to one good* (EF1) if for every i, j with $A_j \neq \emptyset$,

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}) \quad \text{for some } g \in A_j.$$

2. The allocation is *envy-free up to any good* (EFX) if for every i, j ,

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}) \quad \text{for every } g \in A_j.$$

An empty bundle is never envied. Removing a good is only a comparison: the actual allocation still contains all the goods. Here EFX uses every good, including zero-valued goods.

EFX $\implies$ EF1.

Two-Player Cut and Choose

Exercise. When can we achieve an EF1 allocation? Try $n = 2$ players, any m goods, and additive valuations. Use the following values to illustrate your rule.

Player	A	B	C
1	10	9	2
2	1	8	4

Player 1 builds two bundles, adding each good to the bundle she currently values less (breaking ties arbitrarily). Player 2 chooses her preferred bundle. For the order A, B, C , the cutter produces $\{A\}$ and $\{B, C\}$, worth 10 and 11 to her. Player 2 chooses $\{B, C\}$, worth 12 to her; player 1 keeps $\{A\}$.

To prove the cutter is protected, call the final higher-valued bundle H and the other L . If H is empty, both are empty. Otherwise let g be the last good placed in H . Immediately before that placement, H was worth at most the then-current L . No later good enters H , and L can only grow, so

$$v_1(H \setminus \{g\}) \leq v_1(L).$$

Thus even if the cutter receives L , removing g from H eliminates her envy. If she receives H , she does not envy. The chooser does not envy because she chooses her favorite. The allocation is EF1.

Round Robin

The Round Robin Mechanism. First, set an arbitrary order of the players, denoted $1, 2, \dots, n$. Until all the goods are allocated, go through the players in this order repeatedly. On player i 's turn, she chooses her most valuable remaining good. Ties may be broken arbitrarily.

Theorem 1. *For nonnegative additive valuations, round robin produces an EF1 allocation.*

On the two-player table above, round robin with player 1 first assigns $\{A, C\}$ to player 1 and $\{B\}$ to player 2. Player 1 compares 12 with 9; player 2 compares 8 with 5. This particular allocation is envy-free. The general guarantee below assumes players choose their most valuable remaining goods, not that this behavior is a dominant strategy.

Let $g_{i,k}$ denote the good picked by player i in round k , when that pick exists. Compare two players i, j , evaluating both players' picks using v_i .

Proof. **Case** $i < j$. Draw the pick order. Which good was still available when i made her k th pick?

Player i chooses before j in each round:

$$g_{i,1} \longrightarrow g_{j,1} \longrightarrow g_{i,2} \longrightarrow g_{j,2} \longrightarrow \dots$$

Every $g_{j,k}$ was available when i chose $g_{i,k}$, so $v_i(g_{i,k}) \geq v_i(g_{j,k})$. Player i has at least as many picks as j . Summing the paired comparisons, and using nonnegativity for any extra pick, gives $v_i(A_i) \geq v_i(A_j)$. There is no envy in this direction.

Case $i > j$. Draw the pick order. Which good should be removed from j 's bundle, and how do the remaining picks compare?

Player j chooses first:

$$g_{j,1} \longrightarrow g_{i,1} \longrightarrow g_{j,2} \longrightarrow g_{i,2} \longrightarrow \dots$$

Remove $g_{j,1}$. For every remaining $g_{j,k+1}$, player i 's k th pick exists and comes earlier, so

$$v_i(g_{i,k}) \geq v_i(g_{j,k+1}).$$

Summing gives $v_i(A_i) \geq v_i(A_j \setminus \{g_{j,1}\})$. This also handles an incomplete final round: every remaining pick of j has a preceding pick of i . If A_j is empty there is no envy. Both cases together prove EF1, not envy-freeness of the whole allocation.

How Unfair Can Round Robin Be?

Exercise. There are $n \geq 2$ players with identical additive values: $n - 1$ goods worth n each and n goods worth 1 each. Run round robin in the order $1, \dots, n$. Compare with another allocation. Is the round-robin outcome EFX?

In the first round, players $1, \dots, n-1$ each take a good worth n , and player n takes a good worth 1. The remaining $n-1$ unit goods go to the first $n-1$ players. Thus they receive bundles worth $n+1$ and the last player receives value 1.

Removing a large good from an envied bundle leaves value 1, as required for EF1. But removing its unit good leaves value $n > 1$, so the allocation is not EFX.

Instead, give one large good to each of the first $n-1$ players and all n unit goods to the last player. Every player then receives value n . The total value is n^2 , so this allocation gives everyone exactly their proportional share.

Maximin Share

Suppose a player could partition the goods into n bundles, but had to take her least valuable bundle. What value could she guarantee herself?

Definition 3. The *maximin share* of player i is

$$\text{MMS}_i = \max_{(B_1, \dots, B_n) \in \Pi_n(\mathcal{M})} \min_{1 \leq j \leq n} v_i(B_j),$$

where $\Pi_n(\mathcal{M})$ is the set of partitions into n bundles, possibly empty. An allocation A is *MMS-fair* if $v_i(A_i) \geq \text{MMS}_i$ for every player i .

Whose valuation is used for all n bundles in this definition? How does MMS_i compare with the proportional share $v_i(\mathcal{M})/n$?

All bundles are evaluated using the same player's valuation v_i . For every partition, its least valuable bundle is worth at most the average:

$$\min_j v_i(B_j) \leq \frac{\sum_j v_i(B_j)}{n} = \frac{v_i(\mathcal{M})}{n}.$$

Taking the maximum over partitions preserves this bound, so $\text{MMS}_i \leq v_i(\mathcal{M})/n$. A player's partition is hypothetical: different players may choose different maximizing partitions.

Exercise. In the example with $n-1$ goods worth n and n goods worth 1, compute each player's MMS. What fraction of her MMS does the last player receive under round robin?

The partition into $n-1$ singleton large goods and one bundle of all unit goods has value n in every bundle. Thus $\text{MMS}_i \geq n$. The proportional-share upper bound is $n^2/n = n$, so $\text{MMS}_i = n$. The last player receives value 1 under round robin, only $1/n$ of her MMS. Thus EF1 does not imply MMS-fairness.

Research Questions in Fair Division

The following results give context for the definitions.

Does a fair allocation exist? Round robin guarantees EF1 for any number of additive players. Replacing “some good” by “every good” leads to a much harder question: does a complete EFX allocation always exist? The unrestricted additive case remains open, including four players with arbitrarily many goods. EFX is known for three players, and more generally for any number of players with at most three distinct additive valuations [HV et al., 2025]. A 2026 preprint establishes four-player existence for up to nine goods [Alkassar et al., 2026]. Restricting the number of valuation types or goods gives precise intermediate questions.

If exact fairness fails, what fraction can everyone receive? Exact MMS allocations need not exist for additive valuations. The round-robin example only shows that one algorithm can fail; MMS impossibility means that *every* allocation fails on some instance. An α -MMS allocation satisfies $v_i(A_i) \geq \alpha \text{MMS}_i$ for every i . A recent result guarantees $\alpha = 10/13$; for each fixed $\varepsilon > 0$, a $(10/13 - \varepsilon)$ -MMS allocation can be found in polynomial time [Heidari et al., 2025]. Determining the optimal universal factor remains open. Existence and efficient computation are separate questions.

Can fairness coexist with efficiency? For additive goods, EF1 and Pareto optimality can be achieved together. Maximum Nash welfare supplies such an allocation [Caragiannis et al., 2019]. When all players can receive positive value, this means maximizing $\prod_i v_i(A_i)$. If this is impossible, first maximize the number receiving positive value, then maximize their product. Exact optimization is computationally hard. This raises algorithmic questions about obtaining fairness and efficiency together, not merely finding a fair outcome.

Which assumptions matter? Additivity is consequential: a 2026 preprint gives three-player, eight-good counterexamples to EFX for submodular valuations, which permit diminishing marginal values [Mackenzie and Suzuki, 2026]. This does not settle the additive question. Other natural directions change the feasible bundles, the players’ entitlements, or whether values are reported strategically. Each change requires specifying which fairness benchmark and guarantee should survive.

Rent division changes a different assumption: money is available. Prices can compensate for assigning different indivisible rooms.

Rent Division

Housemates disagree about which room is best. Prices let us compensate the housemate assigned a less desirable room. An envy-free outcome makes each person prefer their own room at its price to every alternative.

There are n housemates, n rooms, and total rent R . Housemate i values room j at $v_{ij} \geq 0$, with $\sum_j v_{ij} = R$ for each i . Utilities are quasilinear; prices may be negative unless we explicitly require otherwise.

Definition 4. An *outcome* is a pair (σ, p) , where the assignment σ is a bijection from housemates to rooms and the price vector $p = (p_1, \dots, p_n)$ satisfies $\sum_j p_j = R$ (the landlord gets paid, exactly). The utility of housemate i is $u_i = v_{i\sigma(i)} - p_{\sigma(i)}$.

Definition 5. An outcome (σ, p) is *envy-free* if for every housemate i and every room j ,

$$v_{i\sigma(i)} - p_{\sigma(i)} \geq v_{ij} - p_j. \quad (1)$$

That is, nobody wants to trade places with anybody else, where trading places means swapping both the room *and* the rent that comes with it.

Two Rooms

Two housemates have total rent $R = 100$ and the following values:

Housemate	Room A	Room B
1	70	30
2	60	40

Assign room A to housemate 1 and room B to housemate 2. Let $p_A = q$, so $p_B = 100 - q$. For which prices is this assignment envy-free?

Housemate 1 does not envy when

$$70 - q \geq 30 - (100 - q) = q - 70, \quad \text{or } q \leq 70.$$

Housemate 2 does not envy when

$$40 - (100 - q) = q - 60 \geq 60 - q, \quad \text{or } q \geq 60.$$

Thus the envy-free prices are exactly $60 \leq q \leq 70$. Equal rents, $q = 50$, do not work: housemate 2 would prefer room A at its price. Notice that both housemates value room A more, but suitable prices make each prefer her assigned room.

Fairness and Efficiency

Envy-freeness compares each person's room with all the alternatives at the same prices. Does it also tell us which assignment maximizes total value?

Theorem 2. Let (σ, p) be any envy-free outcome of a rent division instance. Then σ maximizes total value: for every other assignment τ , $\sum_i v_{i\sigma(i)} \geq \sum_i v_{i\tau(i)}$.

Fix any alternative assignment τ , keeping the price of each room fixed. Compare housemate i 's assigned room with $\tau(i)$, then sum over housemates.

Proof. Let τ be any assignment. Apply the envy inequalities for housemate i with the room $j = \tau(i)$, and sum over all i :

$$\sum_i v_{i\sigma(i)} - \sum_i p_{\sigma(i)} \geq \sum_i v_{i\tau(i)} - \sum_i p_{\tau(i)}.$$

Now σ and τ are both bijections onto the same set of n rooms, so both $\sum_i p_{\sigma(i)}$ and $\sum_i p_{\tau(i)}$ are just $\sum_j p_j = R$. The payment terms cancel. $\square$

Exercise. Give a welfare-maximizing assignment and prices summing to R that are not envy-free.

Let $R = 100$ and let both housemates value both rooms at 50. Every assignment maximizes welfare. At prices $(100, 0)$, the person in room 1 envies room 2.

Choosing within the Interval

For the two-room example, how do the utilities change as q moves from 60 to 70?

q	p_A	p_B	u_1	u_2
60	60	40	10	0
65	65	35	5	5
70	70	30	0	10

All three outcomes are envy-free. The assignment has total value $70 + 40 = 110$, and total rent is 100, so total utility is always 10. Changing the prices redistributes this utility. Envy-freeness alone does not select a unique outcome.

More generally, with two housemates, values normalized by $v_{i1} + v_{i2} = R$, and housemate i assigned room i , the two envy inequalities give

$$R - v_{22} \leq p_1 \leq v_{11}.$$

Indeed, housemate 1's inequality is $v_{11} - p_1 \geq p_1 - v_{11}$, and housemate 2's is $v_{22} - R + p_1 \geq R - v_{22} - p_1$. This interval is nonempty exactly when $v_{11} + v_{22} \geq R$. The other assignment has value $2R - v_{11} - v_{22}$, so this is exactly the condition that the chosen assignment maximizes total value.

The midpoint gives

$$p_1 = \frac{R - v_{22} + v_{11}}{2}, \quad u_1 = u_2 = \frac{v_{11} + v_{22} - R}{2}.$$

With these normalized two-person values, equal utilities are therefore achievable at envy-free prices. This calculation does not assert that equal utilities are feasible in every larger rent-division instance.

Optional: Two-Player MMS

Exercise. With two players and additive valuations, player 1 chooses a partition attaining her MMS, and player 2 chooses her preferred bundle. Prove that this is MMS-fair. Is the earlier greedy EF1 partition necessarily an MMS partition?

Both bundles in player 1's chosen partition are worth at least MMS_1 to her, so the bundle left to her meets her benchmark. Player 2 chooses the better of two bundles, and therefore receives at least $v_2(\mathcal{M})/2 \geq \text{MMS}_2$. Thus both guarantees hold.

The greedy EF1 partition need not attain MMS. If the cutter values three goods at 2, 2, 3 and processes them in that order, greedy balancing produces values 5 and 2. But the partition $\{3\}, \{2, 2\}$

has minimum 3, which is optimal: total value is 7, the average is $7/2$, and bundle values are integral. This is an existence argument using an MMS partition, not a claim that the greedy construction finds one.

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